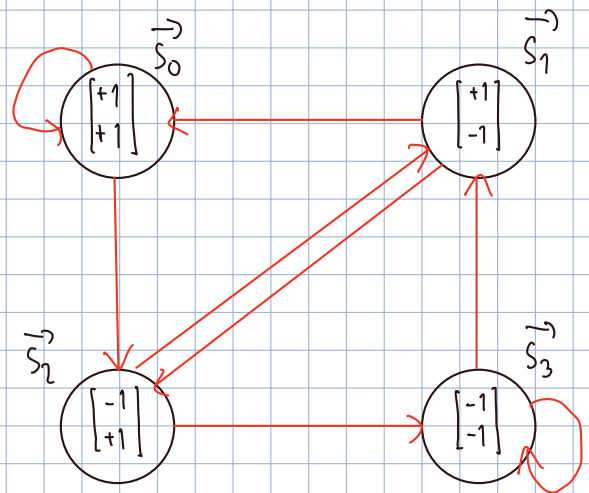


ECE 461 - HW07

1) $y[m] = h[0]x[m] + h[1]x[m-1] + w[m]$ for 2-tap ISI channel

$$y[m] = x[m] - x[m-1] + w[m] \Rightarrow \begin{cases} h[0] = 1 \\ h[1] = -1 \end{cases} \vec{h}[m] = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Defining $\vec{s}[m] = \begin{bmatrix} x[m] \\ x[m-1] \end{bmatrix}$, possible transitions of the finite state machine are sketched below:



ML rule states that out of 2^n paths, we should choose the one with the maximum likelihood

$$\mathcal{L}_{\vec{s}_1} = \prod_{m=1}^n \frac{1}{(2\pi\sigma^2)^{1/2}} e^{-\frac{(y[m] - \vec{h}^T \vec{s}[m])^2}{2\sigma^2}}$$

$$\hat{\vec{s}}_{ML} = \underset{\vec{s}}{\operatorname{argmin}} \|\vec{y} - \vec{h}^T \vec{s}\|^2 = \underset{\vec{s}}{\operatorname{argmin}} \sum_{m=1}^n (y[m] - \vec{h}^T \vec{s}[m])^2$$

But according to Viterbi algorithm, we can achieve this by just choosing lower cost path $(y[m] - \vec{h}^T \vec{s}[m])^2$ at each time-step.

Let us calculate $C_i[m] = \left(y[m] - [1 \ -1]^T \cdot \vec{s}_i \right)^2$, the cost of going to i -th state at time m , from one of the 2 possible previous states (for example, one can arrive at $\vec{s}_2$ from either $\vec{s}_0$ or $\vec{s}_1$)

First time step ($m=1$)

$$C_0[1] = (0.5 - 0)^2 = 0.25$$

$$C_1[1] = (0.5 - 2)^2 = 2.25$$

$$C_2[1] = (0.5 - (-2))^2 = 6.25$$

$$C_3[1] = (0.5 - 0)^2 = 0.25$$

Second time step ($m=2$)

$$C_0[2] = (0.5 - 0)^2 = 0.25$$

$$C_1[2] = (0.5 - 2)^2 = 2.25$$

$$C_2[2] = (0.5 - (-2))^2 = 6.25$$

$$C_3[2] = (0.5 - 0)^2 = 0.25$$

Third time step ($m=3$)

$$C_0[3] = (3 - 0)^2 = 9$$

$$C_1[3] = (3 - 2)^2 = 1$$

$$C_2[3] = (3 - (-2))^2 = 25$$

$$C_3[3] = (3 - 0)^2 = 9$$

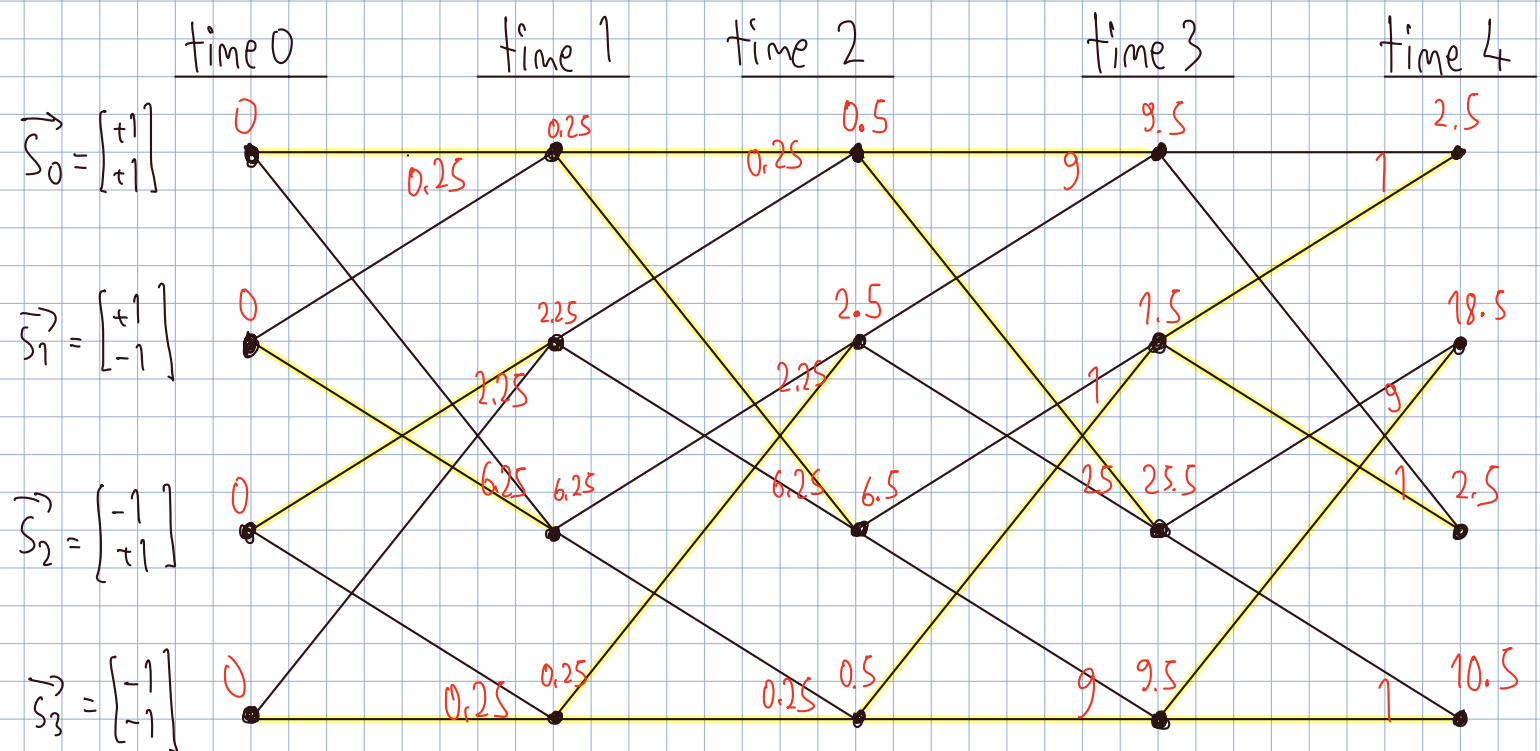
Fourth time step ($m=4$)

$$C_0[4] = (-1 - 0)^2 = 1$$

$$C_1[4] = (-1 - 2)^2 = 9$$

$$C_2[4] = (-1 - (-2))^2 = 1$$

$$C_3[4] = (-1 - 0)^2 = 1$$



b) To find ML estimates, we look at minimum cumulative cost at $n=4$. Minimum cost 2.5 is tied between states s_0 and s_2 . We must trace both path back

Possible states

ML estimates

$s_0 \rightarrow s_1 \rightarrow s_3 \rightarrow s_3 \Rightarrow$

$x[4]=+1, x[3]=+1, x[2]=-1, x[1]=-1$

$s_2 \rightarrow s_1 \rightarrow s_3 \rightarrow s_3 \Rightarrow$

$x[4]=-1, x[3]=1, x[2]=-1, x[1]=-1$

backward in time

$\Downarrow$

Both of these sequences are equally likely

$$2) \quad y[m] = x[m] + \frac{1}{3} x[m-1] + \frac{1}{4} x[m-2] + w[m]$$

$$\hookrightarrow x[m] = 0 \quad \text{for } m \neq 1, 2$$

$$\hookrightarrow x[1], x[2] \text{ i.i.d. with mean 0 and variance } E$$

$$\hookrightarrow w[m] \sim \mathcal{N}(0, \sigma^2)$$

$$\begin{aligned} 2.a) \quad y[1] &= x[1] + w[1] \\ y[2] &= x[2] + \frac{1}{3} x[1] + w[2] \\ y[3] &= \frac{1}{3} x[2] + \frac{1}{4} x[1] + w[3] \end{aligned}$$

We wish to estimate $x[1]$ using $\hat{y}[1] = \sum_{k=1}^3 c_k y[k]$

$$\hat{y}[1] = x[1] \left(c_1 + \frac{c_2}{3} + \frac{c_3}{4} \right) + x[2] \left(c_2 + \frac{c_3}{3} \right) + \underbrace{c_1 w[1] + c_2 w[2] + c_3 w[3]}_{\tilde{w}}$$

$\rightarrow$ In order to eliminate the interference term $x[2]$, we must have:

$$c_2 + \frac{c_3}{3} = 0 \quad \Rightarrow \quad \boxed{c_3 = -3c_2}$$

$\rightarrow$ There is no constraint for c_1 , in terms of ISI cancellation.

$$\begin{aligned} b) \quad \hat{y}[1] &= x[1] \left(c_1 + \frac{c_2}{3} - \frac{3c_2}{4} \right) + \tilde{w} \\ &= x[1] \left(c_1 - \frac{5c_2}{12} \right) + \tilde{w} \end{aligned}$$

$$\text{with } \tilde{w} = c_1 w[1] + c_2 w[2] - 3c_2 w[3]$$

$$E[\tilde{w}^2] = c_1^2 E[w^2[1]] + c_2^2 E[w^2[2]] + 9c_2^2 E[w^2[3]]$$

$$= [c_1^2 + c_2^2 + 9c_2^2] \sigma^2 \quad \text{since iid Gaussian}$$

→ There is no interference term and SINR is basically

$$\text{SINR} = \frac{\mathbb{E} \left[x_{c1}^2 \left(c_1 - \frac{5c_2}{12} \right)^2 \right]}{0 + \mathbb{E} [\tilde{w}^2]} = \frac{\mathbb{E} \left(c_1 - \frac{5c_2}{12} \right)^2}{(c_1^2 + c_2^2 + 9c_2^2) \sigma^2}$$

$$\boxed{\text{SINR} = \frac{\left(c_1 - \frac{5c_2}{12} \right)^2}{c_1^2 + 10c_2^2} \text{SNR}}$$

c) Notice that defining $\vec{h} = [1 \quad 1/3 \quad 1/4]^T$ and $\vec{c} = [c_1 \quad c_2 \quad -3c_2]^T$

$$\vec{h}^T \cdot \vec{c} = \left(c_1 - \frac{5c_2}{9} \right), \quad \|\vec{c}\|_2^2 = c_1^2 + c_2^2 + 9c_2^2$$

$$\text{SINR} = \frac{(\vec{h}^T \cdot \vec{c})^2}{\|\vec{c}\|_2^2} \text{SNR}$$

Using Cauchy-Schwarz Ineq: $(\vec{h}^T \cdot \vec{c})^2 \leq \|\vec{c}\|^2 \cdot \|\vec{h}\|^2$

$$\text{SINR} \leq \|\vec{h}\|^2 = 1^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{4}\right)^2 = \frac{169}{144}$$

* ∇_0 HOWEVER, note that to achieve maximum possible SINR we need $\vec{c} = \alpha \vec{h}$ which violates the zero interference constraint since $c_3 \neq -3c_2$ anymore.

Thus we need to find $\vec{u}, \vec{v} \in \mathbb{R}^2$, such that

$$\left. \begin{array}{l} \vec{u} \cdot \vec{v} = c_1 - \frac{5}{12} c_2 \\ \|\vec{u}\|^2 = c_1^2 + 10 c_2^2 \end{array} \right\} \begin{array}{l} \vec{u} = [c_1 \quad \sqrt{10} c_2]^T \\ \vec{v} = \left[1 \quad -\frac{5}{12\sqrt{10}} \right]^T \end{array}$$

Then again, using Cauchy-Schwarz

$$\text{SINR} = \frac{(\vec{u} \cdot \vec{v})^2}{\|\vec{u}\|^2} \text{SNR} \leq \|\vec{v}\|^2 \text{SNR} = \left(1 + \frac{25}{1440} \right) \text{SNR}$$

$$\boxed{\text{SINR}_{\max} = \frac{293}{288} \text{SNR}}$$

$$3) \quad y[m] = x[m] + 2x[m-1] + w[m]$$

$$\hookrightarrow x[m] = 0 \quad \text{for } m \neq 1, 2$$

$$\hookrightarrow x[1], x[2] \text{ iid. with mean } 0 \text{ and variance } E$$

$$\hookrightarrow w[m] \sim \mathcal{N}(0, \sigma^2)$$

$$y[1] = x[1] + w[1]$$

$$y[2] = x[2] + 2x[1] + w[2]$$

$$\hat{y}[1] = (c_1 + 2c_2)x[1] + c_2x[2] + c_1w[1] + c_2w[2]$$

$\rightarrow$ To detect $x[1]$ using $\hat{y}[1]$, interference term $x[2]$ must be eliminated.

$$\boxed{c_2 = 0, \text{ no constraint on } c_1} \Rightarrow \text{except } c_1 \neq 0 \text{ trivially.}$$

$$\hat{y}[1] = c_1x[1] + c_1w[1]$$

$$\text{SNR} = \frac{\mathbb{E}[c_1^2 x[1]^2]}{0 + \mathbb{E}[c_1^2 w[1]^2]} = \frac{c_1^2 E}{c_1^2 \sigma^2} = \boxed{\text{SNR}}$$

$\rightarrow$ We are forced to throw away the stronger signal component $y[2]$ which contains a stronger copy of $x[1]$ than $y[1]$ does. This is a weakness of zero-forcing filter!